

Proceeding as usual after row and column operations, we get the following optimum assignment :

$1 \rightarrow D, 2 \rightarrow B, 3 \rightarrow C$ and $4 \rightarrow E$.

The maximum profit according to this assignment is
 $\text{Rs. } (10.10 + 8.40 + 11.10 + 8.00)$
 $= \text{Rs. } 37.60.$

EXAMPLES IX (A)

Find the optimal assignments to find the minimum cost for the assignment problems with the following cost matrices (1 - 10) :

01

	J_1	J_2	J_3
P_1	12	24	15
P_2	23	18	24
P_3	30	14	28

[Ans.] $P_1 \rightarrow J_1, P_2 \rightarrow J_3, P_3 \rightarrow J_2$;
 Min. cost = 50.]

02

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

[Kalyani Hons., 1994]

[Ans.] $A \rightarrow 2, B \rightarrow 3, C \rightarrow 4, D \rightarrow 1$;
 Min. cost = 38.]

03

	1	2	3	4
1	1	4	6	3
2	9	7	10	9
3	4	5	11	7
4	8	7	8	5

[Ans.] $1 \rightarrow 1, 2 \rightarrow 3, 3 \rightarrow 2, 4 \rightarrow 4$;
 Min. cost = 21.]

04

	I	II	III	IV	V
A	6	5	8	11	16
B	1	13	16	1	10
C	16	11	8	8	8
D	9	14	12	10	16
E	10	13	11	8	16

[Ans.] $A \rightarrow II, B \rightarrow I, C \rightarrow V, D \rightarrow III, E \rightarrow IV$;
 or $A \rightarrow II, B \rightarrow IV, C \rightarrow V, D \rightarrow I, E \rightarrow III$;
 Min. cost = 34.]

05

	I	II	III	IV
A	5	3	1	8
B	7	9	2	6
C	6	4	5	7
D	5	7	7	6

[Ans.] $A \rightarrow III, B \rightarrow IV,$
 $C \rightarrow II, D \rightarrow I$;
 Min. cost = 16.]

06

	M_1	M_2	M_3	M_4
J_1	10	24	30	15
J_2	16	22	28	12
J_3	12	20	32	10
J_4	9	26	34	16

[Ans.] $J_1 \rightarrow M_3, J_2 \rightarrow M_2, J_3 \rightarrow M_4, J_4 \rightarrow M_1$;
 $J_1 \rightarrow M_3, J_2 \rightarrow M_4, J_3 \rightarrow M_2, J_4 \rightarrow M_1$;
 $J_1 \rightarrow M_2, J_2 \rightarrow M_3, J_3 \rightarrow M_4, J_4 \rightarrow M_1$;
 Min. cost = 71.]



07

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
1	11	17	8	16	20
2	9	7	12	6	15
3	13	16	15	12	16
4	21	24	17	28	26
5	14	10	12	11	15

[Ans. 1 → *a*, 2 → *d*, 3 → *e*,
4 → *c*, 5 → *b*; Min. cost = 60.]

08

	MEN				
	I	II	III	IV	V
A	1	3	2	3	6
B	2	4	3	1	5
C	5	6	3	4	6
D	3	1	4	2	2
E	1	5	6	5	4

[Calcutta Hons., 2006]

[Ans. A → I, B → IV, C → III,
D → II, E → V;
or A → II, B → IV, C → III,
D → V, E → I;
Min. cost = 10.]

09

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
1	2	9	2	7	1
2	6	8	7	6	1
3	4	6	5	3	1
4	4	2	7	3	1
5	5	3	9	5	1

[North Bengal Hons., 2007]

[Ans. 1 → *c*, 2 → *e*, 3 → *a*, 4 → *d*, 5 → *b*;
1 → *c*, 2 → *e*, 3 → *d*, 4 → *b*, 5 → *a*;
1 → *c*, 2 → *e*, 3 → *d*, 4 → *a*, 5 → *b*;
Min. cost = 13.]

10

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
1	18	26	17	11
2	13	28	14	26
3	38	19	18	15
4	19	26	24	10

[Calcutta Hons., 1987, 2002]

[Ans. 1 → *c*, 2 → *a*, 3 → *b*, 4 → *d*;
Min. cost = 59.]

- 11 One car is available at each of the stations 1, 2, 3, 4, 5, 6 and one car is required at each of the stations 7, 8, 9, 10, 11, 12. The distances between the various stations are given in the adjacent matrix. How should the cars be despatched so as to minimize the total mileage coverage?

	7	8	9	10	11	12
1	41	72	39	52	25	51
2	22	29	49	65	81	50
3	27	39	60	51	32	32
4	45	50	48	52	37	43
5	29	40	39	26	30	33
6	82	40	40	60	51	30

[Ans. 1 → 11, 2 → 8, 3 → 7,
4 → 9, 5 → 10, 6 → 12;
Minimum mileage = 185.]

- 12 A team of five horses and five riders has entered a jumping show contest.

	RIDERS				
	<i>R</i> ₁	<i>R</i> ₂	<i>R</i> ₃	<i>R</i> ₄	<i>R</i> ₅
H ₁	5	3	4	7	1
H ₂	2	3	7	6	5
H ₃	4	1	5	2	4
H ₄	6	8	1	2	3
H ₅	4	2	5	7	1

The number of penalty points to be expected when each rider rides any horse is given in the adjacent tableau. How should the horses be allotted to the riders so as to minimize the expected loss of the team?

[Ans. H₁ → *R*₅, H₂ → *R*₁, H₃ → *R*₄,
H₄ → *R*₃, H₅ → *R*₂;
Min. loss = 8.]

13 Use Hungarian method to solve the following assignment problem :

PROJECT	LOCATION				
	I	II	III	IV	V
A	15	21	6	4	9
B	3	40	21	10	7
C	9	6	5	8	10
D	14	8	6	9	3
E	21	16	18	7	4

[Ans.] $A \rightarrow IV, B \rightarrow I, C \rightarrow II, D \rightarrow III, E \rightarrow V$; Min. cost = 23.]

14 A computer centre has three expert programmers. The centre needs three application programmes to be developed. The head of the computer centre, after studying carefully the programmes to be developed, estimates the computer time in minutes required by the experts to the application programmes as follows :

		PROGRAMMES		
		A	B	C
PROGRAMMERS	1	120	100	80
	2	80	90	110
	3	110	140	120

Assign the programmers to the programmes in such a way that the total computer time is least.

[Kalyani Hons., 1988]

[Ans.] $1 \rightarrow C, 2 \rightarrow B, 3 \rightarrow A$; Min. time = 280 minutes.]

15 A car hire company has one car at each of the five depots a, b, c, d and e . A customer in each of the five towns A, B, C, D and E requires a car. The distance (in miles) between the depots (origins) and the towns (destinations) where the customers are, is given in the adjacent distance matrix.

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	125	170	185
D	50	50	80	80	110
E	55	35	80	80	105

How should the cars be assigned to the customers so as to minimize the distance travelled.

[Calcutta M. Sc., 1976]

[Ans.] $A \rightarrow b, B \rightarrow e, C \rightarrow c, D \rightarrow a, E \rightarrow d$; Min. dist. = 560 miles.]

16 A company has four machines on which to do three jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the adjacent tableau. What are the job assignment which will minimize the cost?

		MACHINES			
		W	X	Y	Z
JOBS	A	18	24	28	32
	B	8	13	17	19
	C	10	15	19	22

[Add a fictitious row to balance the problem.]

[Ans.] $A \rightarrow W, B \rightarrow X, C \rightarrow Y$;
or $A \rightarrow W, B \rightarrow Y, C \rightarrow X$; Min. Cost = 50.]

[Kalyani M. Sc., 1983]



17 A company is faced with the problem of assigning six different machines to five different jobs. The costs are estimated in the adjacent table (hundred of rupees)

Solve the problem assuming that the objective is to minimize the total cost.

[Jadavpur M. Sc., 1995]

[Ans] $1 \rightarrow 4, 2 \rightarrow 2, 3 \rightarrow 5, 4 \rightarrow 3, 5 \rightarrow 1;$

or $1 \rightarrow 4, 2 \rightarrow 1, 3 \rightarrow 5, 4 \rightarrow 3, 5 \rightarrow 2;$ Min. cost = 19.]

		JOBS				
		1	2	3	4	5
MACHINES	1	2.5	5	1	5	1
	2	2	5	1.5	7	3
	3	3	6.5	2	8	3
	4	3.5	7	2	9	4.5
	5	4	7	3	9	6
	6	6	9	5	10	6

18 (a) Find the optimal assignment to find the maximum profit for the assignment problem with the adjacent profit matrix.

	I	II	III	IV
A	7	5	4	3
B	8	2	6	4
C	5	3	2	1
D	5	4	1	8

[Ans] $A \rightarrow II, B \rightarrow III, C \rightarrow I, D \rightarrow IV;$
Max. profit = 24.]

(b) The owner of a small machine shop has four machinists available to assign to jobs for the day. Five jobs are offered with expected profit for each machinist on each job as given in the adjacent tableau.

	A	B	C	D	E
1	62	78	50	101	82
2	71	84	61	73	59
3	87	92	111	71	81
4	48	64	87	77	80

Find the assignment of machinists to jobs that will result in a maximum profit.

[Ans] $1 \rightarrow D, 2 \rightarrow B, 3 \rightarrow C, 4 \rightarrow E;$ Max. profit = 376.]

(c) A company has five jobs (1, 2, 3, 4, 5) to be done. The adjacent matrix shows the return in rupees on assigning the jobs to the machines a, b, c, d, e . Assign the five jobs to the machines to maximize the expected return.

	a	b	c	d	e
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

[Ans] $1 \rightarrow b, 2 \rightarrow a, 3 \rightarrow e, 4 \rightarrow c, 5 \rightarrow d;$ Max. profit = 191.]

19 Find the minimum cost solution for the 4×4 assignment problem whose cost co-efficients are as given in the adjacent table.

[Add 2 to all the elements and then proceed as usual.]

[Ans] $1 \rightarrow IV, 2 \rightarrow III, 3 \rightarrow II, 4 \rightarrow I;$ Min. cost = -3.]

	I	II	III	IV
1	2	-1	-1	-2
2	1	0	-2	-1
3	1	-1	-2	0
4	2	2	1	1

20 Find the minimum cost solution for the 5×5 assignment problem whose cost co-efficients are as given in the adjacent table.

[Kharagpur, 1987]

[Ans] $1 \rightarrow 3, 2 \rightarrow 2, 3 \rightarrow 5,$

$4 \rightarrow 1, 5 \rightarrow 4$

or $1 \rightarrow 4, 2 \rightarrow 2, 3 \rightarrow 3,$

$4 \rightarrow 5, 5 \rightarrow 1;$

Min. cost = -36.]

	1	2	3	4	5
1	-2	-4	-8	-6	-1
2	0	-9	-5	-5	-4
3	-3	-8	-9	-2	-6
4	-4	-3	-1	0	-3
5	-9	-5	-8	-9	-5

21 Consider the problem of assigning five operators to five machines. The assignment costs in rupees are given in the adjacent table. Operator *B* cannot be assigned to machine 2 and operator *E* cannot be assigned to machine 4. Find the optimal cost of assignment.

[North Bengal Hons., 2006]

Ans. $A \rightarrow 5, B \rightarrow 1, C \rightarrow 4, D \rightarrow 3,$
 $E \rightarrow 2; \text{Min. cost} = 9.]$

	1	2	3	4	5
A	8	4	2	6	1
B	0	—	5	5	4
C	3	8	9	2	6
D	4	3	1	0	3
E	9	5	8	—	5

22 There are five pumps available for developing five wells. The efficiency of each pump in producing the maximum yield at each well is shown in the table given below. In what way the pumps be assigned to the wells so as to maximize the overall efficiency?

WELLS

	I	II	III	IV	V
A	45	40	65	25	55
B	50	30	25	60	30
C	25	20	10	20	40
D	35	25	30	25	20
E	80	60	50	70	50

Ans. $A \rightarrow \text{III}, B \rightarrow \text{IV}, C \rightarrow \text{V}, D \rightarrow \text{II}, E \rightarrow \text{I}.]$

23 Five operators have to be assigned to five machines.

The assignment costs are given below.

Operator *A* cannot operate machine III and operator *C* cannot operate machine IV. Find the optimal assignment schedule.

MACHINES

	I	II	III	IV	V
A	5	5	—	2	6
B	7	4	2	3	4
C	9	3	5	—	3
D	7	2	6	7	2
E	6	5	7	9	1

Ans. $A \rightarrow \text{IV}, B \rightarrow \text{III}, C \rightarrow \text{V}, D \rightarrow \text{II}, E \rightarrow \text{I}$

or $A \rightarrow \text{IV}, B \rightarrow \text{III}, C \rightarrow \text{II}, D \rightarrow \text{I}, E \rightarrow \text{V}; \text{Min. cost } 15.]$

24 A company has four territories open and four salesmen available for assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales :

Territory	I	II	III	IV
Annual sales (Rs.)	60,000	50,000	40,000	30,000

The four salesmen are also considered to differ in ability. It is estimated that working under the same condition their yearly sales would be proportionately as follows :

Salesman	A	B	C	D
Proportion	7	5	5	4



If the criterion be maximum expected total sales, the intuitive answer is to assign the best salesman to the richest territory, the next best salesman to the second richest and so on. Verify this answer by the assignment technique.

[Kalyani M. Sc., 1995]

[To avoid fractional values of annual sales of the salesman, we find the effectiveness matrix by considering the sales for 21 years which is the sum of the proportions $7 + 5 + 5 + 4$. Taking Rs. 10,000 as one unit, we get the maximum sale matrix as

		6	5	4	3
		I	II	III	IV
Territories → Sales proportion	7 A	42	35	28	21
	5 B	30	25	20	15
	5 C	30	25	20	15
	4 D	24	20	16	12

Convert this into a minimization problem and then proceed as usual.]

[Ans.] Possible assignments are

(i) $A \rightarrow I, B \rightarrow II, C \rightarrow III, D \rightarrow IV$;

(ii) $A \rightarrow I, B \rightarrow III, C \rightarrow II, D \rightarrow IV$.]

25 An air-line that operates seven days a week has time table shown below. Crews must have a minimum layover of 5 hours between flights. Obtain the pairing of flights that minimizes the layover time away from home. For any given pairing, the crew will be based at the city that results in the smaller layover.

Delhi-Jaipur			Jaipur-Delhi		
Flight No.	Depart	Arrive	Flight No.	Depart	Arrive
1	7.00 A.M.	8.00 A.M.	101	8.00 A.M.	9.15 A.M.
2	8.00 A.M.	9.00 A.M.	102	8.30 A.M.	9.45 A.M.
3	1.30 P.M.	2.30 P.M.	103	12.00 Noon	1.15 P.M.
4	6.30 P.M.	7.30 P.M.	104	5.30 P.M.	6.45 P.M.

For each pair, also mention the town where the crew should be based.

[Agra M. Stat., 1974]

[Construct tables I and II for times of layover between the flights (Table I : when the crew is based in Delhi and Table II : when the crew is based in Jaipur). Take 15 minutes for one unit. The layover time between flights 1 and 101 is 24 hours = 96 units between 8 A.M. and 8 A.M. and so on.]

Table I

	101	102	103	104
1	96	98	112	38
2	92	94	108	34
3	70	72	86	108
4	50	52	66	88

Table II

	101	102	103	104
1	87	85	71	49
2	91	89	75	53
3	113	111	97	75
4	37	35	21	95

Then construct the table for the smallest layover times between flights from Tables I and II. Mark the layover times with asterics to indicate the crew-base at Jaipur otherwise the crew-base will be at Delhi. Thus the table showing the minimum layover times will be

	101	102	103	104
1	87*	85*	71*	38
2	91*	89*	75*	34
3	70	72	86	75*
4	37*	35*	21*	88

Then apply assignment technique to find the optimal assignment.]

[Ans.] (i) $3 \rightarrow 101, 4 \rightarrow 102, 1 \rightarrow 103, 2 \rightarrow 104$;

(ii) $3 \rightarrow 101, 1 \rightarrow 102, 4 \rightarrow 103, 2 \rightarrow 104$. Min. layover time 52 h. 30 m.]